

PASSCLAIM--gut health and immunity.

BACKGROUND: The gut and immune system form a complex integrated structure that has evolved to provide effective digestion and defence against ingested toxins and pathogenic bacteria. However, great variation exists in what is considered normal healthy gut and immune function. Thus, whilst it is possible to measure many aspects of digestion and immunity, it is more difficult to interpret the benefits to individuals of variation within what is considered to be a normal range. Nevertheless, it is important to set standards for optimal function for use both by the consumer, industry and those concerned with the public health. The digestive tract is most frequently the object of functional and health claims and a large market already exists for gut-functional foods worldwide. AIM: To define normal function of the gut and immune system and describe available methods of measuring it. RESULTS: We have defined normal bowel habit and transit time, identified their role as risk factors for disease and how they may be measured. Similarly, we have tried to define what is a healthy gut flora in terms of the dominant genera and their metabolism and listed the many, varied and novel methods for determining these parameters. It has proved less easy to provide boundaries for what constitutes optimal or improved gastric emptying, gut motility, nutrient and water absorption and the function of organs such as the liver, gallbladder and pancreas. The many tests of these functions are described. We have discussed gastrointestinal well being. Sensations arising from the gut can be both pleasant and unpleasant. However, the characteristics of well being are ill defined and merge imperceptibly from acceptable to unacceptable, a state that is subjective. Nevertheless, we feel this is an important area for future work and method development. The immune system is even more difficult to make quantitative judgements about. When it is defective, then clinical problems ensue, but this is an uncommon state. The innate and adaptive immune systems work synergistically together and comprise many cellular and humoral factors. The adaptive system is extremely sophisticated and between the two arms of immunity there is great redundancy, which provides robust defences. New aspects of immune function are discovered regularly. It is not clear whether immune function can be "improved". Measuring aspects of immune function is possible but there is no one test that will define either the status or functional

capacity of the immune system. Human studies are often limited by the ability to sample only blood or secretions such as saliva but it should be remembered that only 2% of lymphocytes circulate at any given time, which limits interpretation of data. We recommend assessing the functional capacity of the immune system by: measuring specific cell functions ex vivo. measuring in vivo responses to challenge, e. g. change in antibody in blood or response to antigens. determining the incidence and severity of infection in target populations during naturally occurring episodes or in response to attenuated pathogens.